

# System Design Basics

## Lesson 1: Introduction to System Design

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### Introduction to System Design - Study Notes

#### Key Concepts

- What is system design?
- Scope vs. architecture
- Non functional requirements

#### Summary

System design is the process of defining the structure, components, and interactions of a software system to meet both functional and non functional goals. It starts with understanding the problem, defining scope, and identifying constraints such as performance, scalability, and reliability. A well designed system balances trade offs and provides a blueprint for implementation.

#### Important Points

- Clear problem statement drives design decisions.
- Non functional requirements often dictate architecture choices.
- Iterative refinement is key.

#### Example

Designing a URL shortener: identify entities (URL, user), choose storage (NoSQL), and outline API endpoints.

#### Quick Review

- What differentiates system design from coding?
- Why are non functional requirements critical?

